

Ostbayerische Technische Hochschule Amberg-Weiden

Die Fakultät Maschinenbau/Umwelttechnik

LCA on usage of rechargeable and disposable AA batteries in Germany

Group Report

Module: Methods of Life Cycle Assessment

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1. Executive Summary

Study Overview

This study focuses on a life cycle assessment of the use of rechargeable and disposable AA batteries in Germany. It concentrates on a German manufacturer that produces both types of batteries in Germany and is taking into account the mining, manufacturing, use phase and end of life. The impact categories Climate Change, Acidification, Eutrophication, Ozone depletion, Photochemical Oxidation, Human toxicity, Terrestrial Ecotoxicity are taken into account for that.

The functional unit should supply low-power household appliances with 2 W of power for one year. This allows for a fair comparison between rechargeable and non-rechargeable batteries.

Key Results

The results show that raw material extraction and production have the greatest impact on environmental pollution. Rechargeable batteries, however, have an additional impact during charging. With frequent use, the impact per battery decreases with every charge, making them more environmentally friendly overall.

When calculating the impact per defined functional unit, rechargeable batteries exhibit lower overall impacts in all impact categories. The emissions from alkaline batteries are at least 20 times higher (ozone depletion) than those from the required NiMH rechargeable batteries. Considering Eutrophication and Climate Change, we reach values 30 times higher, and for Photochemical oxidation, Human toxicity, and terrestrial ecotoxicity, even more than 40 times higher. The greatest difference, at over 50 times higher, is found in Acidification.

Recommendations

Proper handling is crucial for optimal battery life and minimizes environmental impact. Consumers should therefore use the rechargeable batteries responsibly according to the data sheet to achieve both cost and environmental benefits compared to disposable batteries.

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2. Introduction

In recent years, the use of electronic devices in every household has increased significantly. Most electronic devices require batteries to function, which means that as the use of electronic devices increases, so does the consumption of batteries. These batteries are not entirely environmentally friendly and are repeatedly criticized, mainly because of the raw materials required for their production. The extraction of raw materials and their further processing can lead to high greenhouse gas emissions. In addition, these materials are often only found in certain countries around the world, some of which are struggling with geopolitical and social difficulties. Current forecasts show that by 2030, electric vehicles and energy storage devices in the EU alone will require 18 times as much lithium and 5 times as much cobalt. In this regard, recycling rare materials plays a particularly important role in ensuring security of supply for further production. Another aspect that influences the environmental impact of batteries is the electricity mix used to generate the electricity. (Davies, Christensen, Avelar, & Heidrich, 2024; Mohammad, et al., 2023).

New battery technologies are developing rapidly to address sustainability issues such as resource availability, environmental impact, and scalability. Their portability and ability to store large amounts of energy have increased their importance for the transportation sector and the integration of renewable energies, reducing dependence on fossil fuels. But they also play an important role in small everyday devices such as razors, keyboards, remote controls, and computer mice. Although a wired solution would also be possible compared to electric cars, rechargeable batteries offer additional convenience (Davies, Christensen, Avelar, & Heidrich, 2024; Mohammad, et al., 2023).

According to the Federal Ministry for the Environment, Climate Action, Nature conservation and Nuclear Safety of Germany, more than 60,000 tonnes of portable batteries and accumulators are placed on the market every year (Federal Ministry for the Environment, 2022).

Existing studies primarily focus on battery recycling processes in order to recover and reuse materials. This brings considerable advantages in terms of energy requirements and CO₂ savings. However, it does not directly consider the advantages and disadvantages of rechargeable batteries compared to disposable ones (Hamade, Ayache, Ghanem, & Masri, 2020).

Further studies are already comparing rechargeable and disposable batteries, taking into account consumer decisions based on brand-name products. This is accompanied by different transport routes due to production in different countries, as well as different conditions at the respective production sites (Dolci, Tua, Grosso, & Rigamonti, 2015).

This study builds on this by focusing on a life cycle assessment of the use of rechargeable and disposable AA batteries in Germany. It concentrates on a German manufacturer that produces both types of batteries in Germany. The aim of this study is to compare these batteries and, taking into account a specific energy requirement, to make conclusive statements in favor of rechargeable or non-rechargeable batteries.

3. Goal And Scope Definition

The goal of this Life Cycle Assessment (LCA) is to compare rechargeable AA batteries and non-rechargeable AA batteries used in Germany. AA batteries are broadly used in daily life for example in remote controls, wall clocks, computer mice and toys. Therefore, this size was chosen, as it is one of the most used in Germany (GP Batteries International Limited, 2025; Umwelt Bundesamt, 2024).

The study aims to identify which type of battery has lower overall environmental impacts and Life Cycle Costs during its whole life cycle. This should enable consumers and businesses make better decisions.

Functional Unit

The functional unit should supply low-power household appliances with 2 W of power for one year. This allows for a fair comparison between rechargeable and non-rechargeable batteries.

Scaling

In order to make sound assumptions for the comparison of the two types, the German battery manufacturer *VARTA* was chosen, which manufactures both disposable and rechargeable batteries in Germany. (VARTA AG, 2025)

The *Varta LR06 AA Mignon* was chosen as the disposable battery. It has a Voltage of 1.5 V and a capacity of 2.97Ah. One of these primary alkaline cylindrical batteries has a weight of 23.2g (VARTA Microbattery GmbH, 2025; VARTA Microbattery GmbH, 2025). The total amount of ingredients considered is listed in **Table 2**.

The *VH 2200 AA* was selected as the rechargeable Nickel metal hydride battery. With a Voltage of 1.2 V and a capacity of 2.2 Ah one battery weighs 30.5 g and can be charged over 500 times (Varta Microbattery GmbH, 2015; VARTA Microbattery GmbH, 2025). The materials considered are listed in detail in **Table 3**.

According to the functional unit 2W should be provided over 8760 h, that equals 17.520 Wh. These value were set into context with the Energy amount provided per AA Battery (disposable and rechargeable). The results are shown in **Table 1**.

Table 1: Amount of Batteries

	Demand	Disposable	rechargeable
Energy [Wh]	17520	4,445	2,64
Amount		3933	6637 (charging cycles)
			14 (batteries)

For this LCA two different scaling factors are needed, as it's a comparative study. A Linear Scaling is assumed and depends on the amount of batteries that are needed. Therefore the scaling factor for disposable batteries is 3933 and for rechargeable batteries is 14 (except for the charging cycles).

System Boundaries

For both battery types, the extraction of raw materials and the manufacture of cylindrical cells were taken into account. The manufacturing process also includes the packaging of the batteries. The use phase differs slightly, as only rechargeable batteries require electricity for charging. The German electricity mix was used for this. Since this study aims to highlight the advantages and disadvantages of using the different types in Germany, the use phase defines the foreground system (**Figure 1**). The end phase, including recycling and waste, was also considered.

Human labour, office buildings, and marketing were excluded from the system. The Equipment manufacturing, including the charging device, was also not taken into account. Transport from the production site to the consumer was not considered due to the focus on manufacture and use in Germany. This applies to both battery types almost equally and therefore has no significant impact on costs and environmental impact.

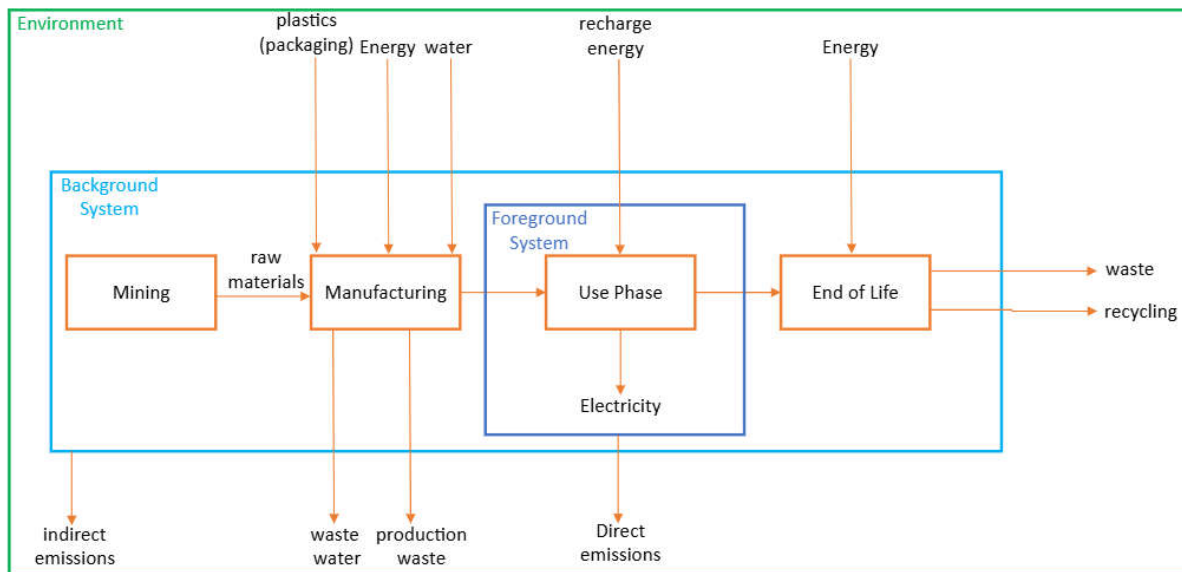


Figure 1: System Boundaries

Of the four processes considered, two are multifunctional. The mining processes consist of several sub-processes with different resources. These processes have been simplified for further modeling. The same applies to the recycling process. Since batteries are composed of many components, there are many outputs, with the largest component being considered the main product.

The specific mining process for potassium hydroxide is examined in detail for a sample mass calculation (see **chapter 4**), as it is used in both batteries.

The following impact categories were taken into account in the further course of the study: Climate Change, Acidification, Eutrophication, Ozone depletion, Photochemical Oxidation, Human toxicity, Terrestrial Ecotoxicity

4. Life Cycle Inventory Analysis

The life cycle assessment model was created using the GaBi software. It illustrates the defined system boundaries. Essentially, the LCA model describes the entire life cycle of rechargeable and non-rechargeable batteries according to the cradle-to-grave methodology. The process begins with the extraction of the raw materials required for the production of both battery types and ends with disposal and material recovery. Key GaBi functions were used to represent process flows, process connections, impact analyses and for calculating results in various impact categories. The required Mining processes are not standard processes but were simplified reconstructed after extensive research.

The following **Table 2** shows the inventory for the disposable batteries. The battery composition was assumed based on the safety data sheet, as this only specifies percentage ranges.

As the use phase is in foreground, not all materials from mining are considered. The direct and indirect emissions shown in figure 1 are considered by GaBi and are included in the resulting impacts in **chapter 5**.

Table 2: Inventory for a disposable AA battery

Flow name	Direction	Category	%	Unit	Amount	Data Quality
Manganese dioxide	Input	Resource	40% ¹	g	9,52	Good
Zinc	Input	Resource	19% ¹	g	4,52	Good
Potassium Hydroxide	Input	Resource	7% ¹	g	1,67	Good
Polymers	Input	Resource	6% ¹	g	1,43	Good
Steel & Nickel	Input	Resource	27% ¹	g	6,43	Good
Water (Manufacturing)	Input	Water	/	m ³	0,000032 ²	Fair
Energy (Mining to Transport)	Input	Energy	/	kWh	0,4822 ³	Fair
Packaging	Input	Resource	/	g	0,4 ²	Good
Energy (End of Life)	Input	Energy	/	kWh	0.07733 ⁴	Fair
Battery	Output	Product	/	pieces	3933 ⁵	Good
Energy Output	Output	Energy	/	kWh	0,004445 ⁵	Good

¹ [Safety Data Sheet](#)

² [Microsoft Word - Absolutely Final MIT LCA Font Change for pdf.doc](#)

³ [\(PDF\) Life Cycle Analysis of AA Alkaline Batteries](#)

⁴ [Estimating energy left in discarded alkaline batteries: Evaluating consumption and recovery opportunities - ScienceDirect](#)

⁵ Calculations, see chapter 3

Waste water	Output	Water emission	/	m ³	0,0000174 ₆	Good
Production Waste	Output	Soil Emission	/	g	1,62 ²	Fair
End of Life Waste	Output	Soil Emission	51% ⁷	g	11,832	Fair
End of Life Recycle	Output	Resource	49% ⁷	g	11,368	Fair

Table 3 below lists the inventory for the rechargeable batteries. The composition was again assumed based on the safety data sheet⁸ and other technical articles. Not only do the production materials differ from disposable batteries, but also the energy required for extraction and production, as well as the recycling process. Additionally, charging energy is essential during the use phase.

Table 3: Inventory for a rechargeable AA battery

Flow name	Direction	Category	%	Unit	Amount	Data Quality
Nickel hydroxide	Input	Resource	27%	g	8,296 ⁹	Good
Mischmetal Nickel Alloy	Input	Resource	24%	g	7,2895 ¹⁰	Good
Steel & Nickel	Input	Resource	20%	g	6,1	Good
Potassium Hydroxide	Input	Resource	15%	g	4,575 ¹¹	Fair
Cobalt & Compounds	Input	Resource	4%	g	1,342 ¹⁰	Good
Polymers	Input	Resource	4%	g	1,22 ¹⁰	Good

⁶ [VARTA_NB_03I.indd](#)

⁷ [Waste statistics - recycling of batteries and accumulators - Statistics Explained - Eurostat](#)

⁸ [Safety Data Sheet](#)

⁹ [\(PDF\) Studies on rechargeable NiMH batteries](#)

¹⁰ [Full article: Characterization of spent nickel–metal hydride batteries and a preliminary economic evaluation of the recovery processes](#)

¹¹ [Nickel Metal Hydride \(NiMH\) Appl Manual Datasheet by Energizer Battery Company | Digi-Key Electronics](#)

Water (manufacturing)	Input	Water	/	m ³	0,005582 ¹²	Fair
Electricity (Battery Charging)	Input	Energy	/	kWh	0,004224 ¹³	Good
Packaging	Input	Resource	/	g	0,4 ²	Good
Energy (Assembly & Transport)	Input	Energy	/	kWh	1,014 ¹⁴	Fair
Energy (End of Life)	Input	Energy	/	kWh	0.166056 ¹⁵	Fair
Battery	Output	Product	/	pieces	14 ⁵	Good
Energy Output	Output	Energy	/	kWh	0,00264 ⁵	Good
Production Waste	Output	Soil Emission	/	g	1,62 ²	Fair
Waste water	Output	Water emission	/	m ³	0,0000174 ⁶	Fair
End of Life Waste	Output	Soil Emission	35% ¹⁰	g	10,675	Fair
End of Life Recycle	Output	Resource	65% ¹⁰	g	19,825	Fair

¹² [Comparative Life Cycle Environmental Impact Analysis of Lithium-Ion \(Lilo\) and Nickel-Metal Hydride \(NiMH\) Batteries](#)

¹³ [DS 55022-2015](#)

¹⁴ [A review of battery life-cycle analysis : state of knowledge and critical needs. \(Technical Report\) | OSTI.GOV](#)

¹⁵ [A-Review-of-Battery-Life-Cycle-Analysis.Argonne-National-Laboratory.pdf](#)

Multi-functionality Implementation

Potassium hydroxide is mainly retrieved out of potassium chloride by the chlor-alkali process. Out of that, the specific masses can be allocated (Edwards, et al., 2017).

Table 4: Supply of potassium hydroxide (KOH) via electrolysis

Material	I/O	Unit	Amount
KCl	Input	kg/kg	1,33
H ₂	Output	kg/kg	0,02
Chlorine	Output	kg/kg	0,63
KOH	Output	kg	1,0

Key Assumptions

Since no specific data on the exact ingredients of a cell is available, the values were assumed according to the safety data sheet for each battery. For rechargeable batteries, 500 charging cycles were assumed, which corresponds to the lower limit of this factor. Both types of batteries supposed to be fully charged when leaving the manufacturing process. Since a German manufacturer was selected, production and use take place in the same area, eliminating the need for shipping and extensive packaging. The emission factor of the German electricity mix was used for the charging energy. Furthermore, actual user behaviour is not modelled accurately, so a ideal use case is modelled.

5. Life Cycle Impact Assessment

For the Life cycle impact assessment the CML-IA method was chosen, as it focuses on Europe and is well established. **Table 5** shows a comparison of the results regarding the impact of disposable and rechargeable batteries. It can be seen that the effects of disposable ones are consistently higher than those of rechargeable ones.

Table 5: Comparative results of the impact assessment

Impact Category	Disposable	Rechargeable	Unit
Climate Change	2203,3	75,37	kg CO ₂ -eq
Acidification	5,31	0,104	kg SO ₂ -eq
Eutrophication	0,596	0,019	kg Phosphate-eq
Ozone depletion	6,42e-11	3,11e-12	kg R11 eq

Photochemical Oxidation	0,343	0,0073	kg Ethene eq
Human toxicity	115	2,65	kg DCB eq
Terrestrial Ecotoxicity	3,52	0,073	kg DCB eq

Results by Life Cycle Stage

The following **Figure 2** shows the percentage share of every process for each impact category, where the bars marked with one represent the rechargeable battery and those marked with two represent the disposable battery.

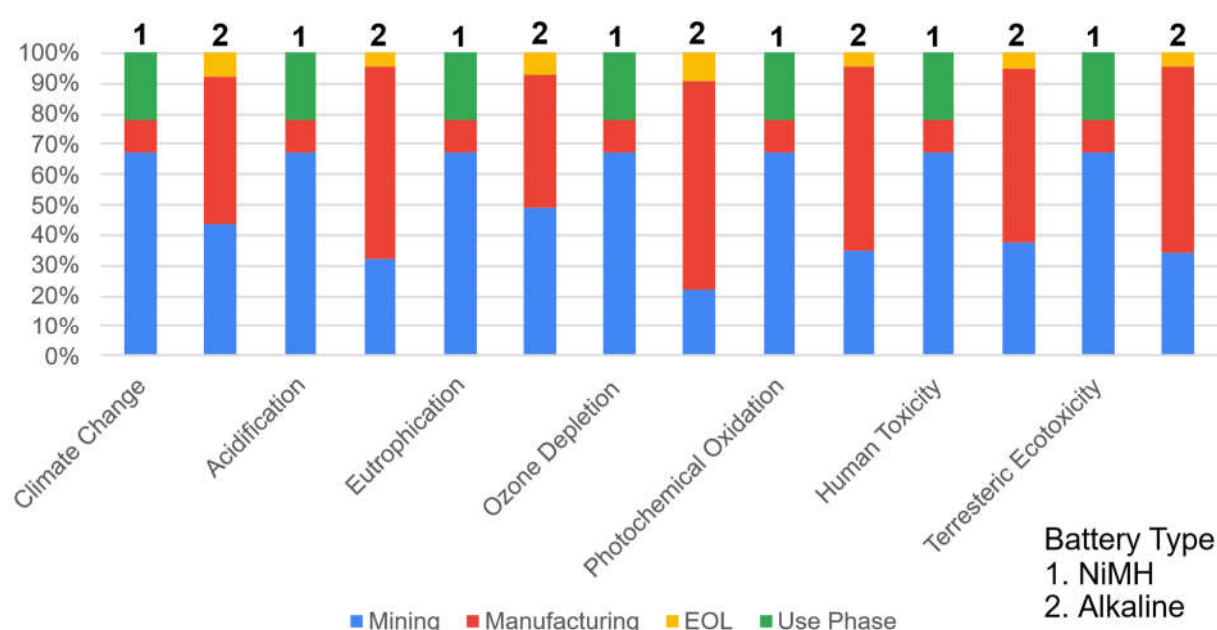


Figure 2: Environmental impacts for each life phase

Environmental Hotspots and Contribution Analysis

Considering the climate change potential, the largest share, approximately 65%, is attributable to the extraction of the raw materials of NiMH batteries, in contrast to alkaline batteries, where the share is only around 45%. However, the share of alkaline battery production is more than four times higher than that of rechargeable batteries.

In view of the impact categories of acidification, eutrophication, ozone depletion, photochemical oxidation, human toxicity and terrestrial ecotoxicity a very similar breakdown emerges when examining NiMH batteries. Approximately 65% of these impacts are consistently attributable to mining, making it by far the most significant factor. Whereas production accounts for approximately 10%, as with climate change.

The usage phase is only affected by the amount of electricity required to recharge the NiMH batteries. It accounts for approximately 20% in all categories.

Looking at the acidification, photochemical oxidation, human toxicity and terrestriec ecotoxicity for alkaline batteries, manufacturing accounts for the largest share around 60%. In contrast, mining typically accounts for approximately 35%. In the case of Eutrophication the mining takes almost 50%, whereas by Ozone depletion only 20 is attributable to the extraction of materials required for Alkaline batteries.

The remaining percentage is due to the end-of-life phase. This accounts for between 5% and 10% of alkaline batteries, while the percentage for NiMH batteries is due to the different recycling processes so small that it is not clearly visible in **Figure 2** (approx. 1%).

6. Life Cycle Interpretation

The results show that raw material extraction and production have the greatest impact on environmental pollution. This is due, on the one hand, to the processes necessary for extracting the raw materials, and on the other hand, to the composition of many different materials used in manufacturing. It should be noted that the impact of mining and manufacturing varies between the two battery types, with the production of alkaline batteries contributing a larger share to the overall impact than that of rechargeable batteries.

For rechargeable batteries, the usage phase also plays a role, as the main impact per battery can be reduced through frequent charging. Therefore, this study focused on the usage phase. Values relating only to production would yield different results. When calculating the impact per defined functional unit, rechargeable batteries show lower overall impacts in all impact categories. The emissions from alkaline batteries are thus 20 to 50 times higher than those from NiMH rechargeable batteries.

The largest share of harmful emissions falls into the category of climate change. The other categories should not be neglected, however, as they have significant environmental impacts despite being present in smaller quantities and are therefore given equal consideration in this study.

Sensitivity Analysis

Consequently, the influence of changes in certain inputs was investigated. For this purpose, the energy consumption during battery production was considered. **Figure 3**

and **4** show the changes in global warming potential (GWP) with a fluctuation of $\pm 10\%$ in the required energy. In both cases, a lower GWP can be observed with reduced production energy. For disposable batteries (**Figure 3**), the global warming potential (GWP) also changes by $\pm 10\%$, whereas for rechargeable batteries, a reduction only leads to a CO₂ saving of 1%, while a 10% increase in electricity consumption results in an 11% increase in CO₂ emissions. This explains the small number of rechargeable batteries required compared to disposable batteries. Increasing efficiency through new/modified processes and machines is therefore a good approach to reducing the ecological footprint of batteries.

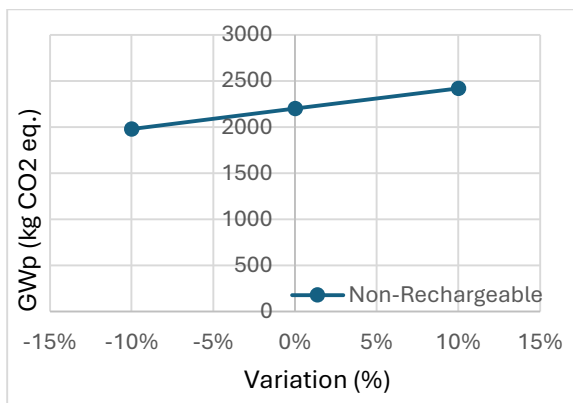


Figure 3: electricity for manufacturing Alkaline

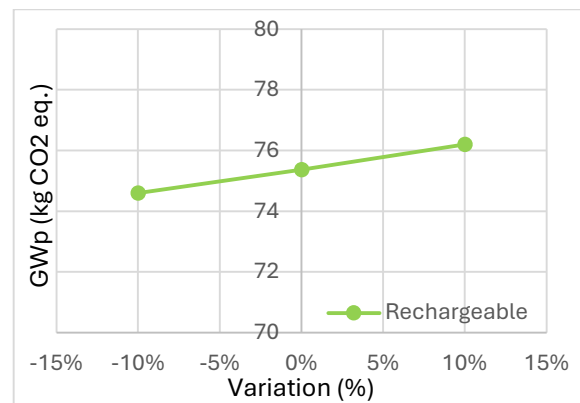


Figure 4: electricity for manufacturing NiMH

Furthermore, a consumer-driven factor was examined. The assumed 500 charging cycles per battery cell can only be achieved with proper handling according to the manufacturer's specifications. Accordingly, **Figure 5** shows the GWP as a function of the required battery quantity. With particularly good handling, the climate change impact can be reduced by 13.5% by using only 10 batteries (-29% in quantity, ~700 cycles per battery). With improper use, and thus 18 batteries required (+29% in quantity, ~400 cycles per battery), the GWP increases by 17.9%. This influence can only be investigated for rechargeable batteries.

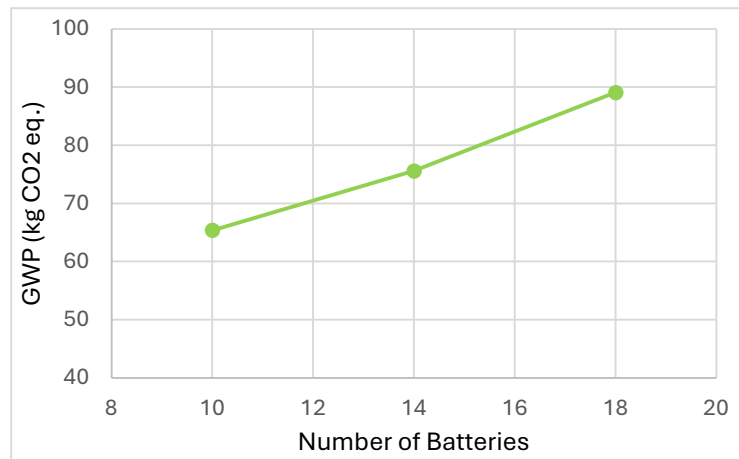


Figure 5: Number of rechargeable batteries required

As a final factor, the end-of-life phase was examined to determine the changes resulting from a higher recycling rate. Since this study does not investigate a closed-loop system but rather a linear model, the software calculates results based on energy supplied to the end-of-life phase, thus showing no influence on the impact categories. A change alone therefore does not constitute an impact as long as the material is not reused. Returning the recycled material to the production process would definitely have an impact, as we know from the scenario analysis (see below) that, for example, nickel then has a lower CO2 footprint. Since this is not included in the model principle, the precise influence was not investigated.

Uncertainty Analysis and Study Limitations

Many of the sources for inventory data are only partially reliable because they do not refer to AA batteries or the exact same type. However, they can serve as a suitable basis for more precise assumptions. The aim of this study was to demonstrate the different impacts of the two investigations. For this reason, both models were modeled with the same level of accuracy. The focus was on the usage phase in order to draw conclusions for consumers.

For the production of the batteries, only "new" materials were considered, no parts from recycled material, which does not reflect reality and affects their environmental impact. Additionally, more materials could be included in the mining and production process, as due to time constraints, the focus was only on larger components (>1%). The study cannot accurately represent the behavior of each consumer; it is up to the individual to handle/use the batteries in the best possible way.

Scenario Analysis

This scenario explains how to select a material from rechargeable and non-rechargeable batteries for recycling in order to achieve the greatest impact on the environment.

For rechargeable batteries, nickel hydroxide is selected because it makes up a large part of the battery, causes a lot of CO₂, and can be reused to manufacture batteries after successful recycling. If enough batteries are collected, for example through government programs, the demand for new nickel decreases, resulting in less mining, less energy consumption, and less pollution (World Steel Association AISBL, 2025; International Energy Agency, 2024).

For non-rechargeable batteries, steel is selected. This is the main metal in the battery casing and is easy to recycle. In addition, recycled steel has a significantly lower carbon footprint. By recycling steel, we can save more than half of the new steel required for each battery (World Steel Association AISBL, 2025).

Ultimately, the scenario shows that recycling works best when we focus on materials that are used frequently, cause a lot of pollution when new, and are easy to recycle in practice, not just in theory. Detailed information on the scenario can be found in the appendix.

7. Conclusions And Recommendations

The results show that the main environmental impact of batteries occurs during their production, particularly during the mining and processing of metals. The greatest damage across all impact categories considered is caused by raw material extraction and manufacturing, not during use. Rechargeable batteries, however, have an additional impact during charging. With frequent use, the impact per battery decreases with every charge, making them more environmentally friendly overall. The results presented are consistent with the results of comparable studies, which were also used for the initial assumptions (Hamade, Ayache, Ghanem, & Masri, 2020; Dolci, Tua, Grosso, & Rigamonti, 2015).

Proper handling is crucial for optimal battery life and minimizes environmental impact. As **Figure 5** clearly shows, the consumer can directly save CO₂ through their behavior, so that with very good handling and only 10 instead of 14 batteries required, ~10 kg

CO₂ can already be saved. Consumers should therefore use them responsibly according to the data sheet to achieve both cost and environmental benefits.

Rechargeable batteries contain valuable materials (e.g., lithium, nickel, cobalt), which makes them more attractive for collection and recycling, especially since the end of life phase has a lower environmental impact than raw material extraction (**Figure 2**). Therefore, collection programs and further research in the field of recycling are necessary.

This study does not indicate when the use of rechargeable batteries becomes more advantageous or how many charging cycles are required to achieve an advantage over disposable batteries, as proper use was assumed. The results could also be applied in other countries. However, the changing impacts due to different CO₂ footprints, etc., in electricity generation should be taken into account. This would affect not only usage but also production (if the production location changes).

Improvement Recommendations

As already mentioned, this study is based on a linear model (see **Figure 1**). To better represent real-world conditions, a circular flow model that incorporates recycled materials proportionally into the production of new batteries would be more suitable for further investigations. Furthermore, additional materials can be integrated during mining, and the use of co-products can be further refined. The same applies to the end-of-life phase.

Future research could therefore not only include recycled materials but, in the context of the energy transition, also consider the influence of clean energy, thus providing consumers and producers with an incentive to switch to renewable energy sources.

8. Individual Contribution Statement

Student Name: Lena Schecklmann

- Main responsibilities: report, summarizing retrieved data
- Specific contributions: summed up everything in the report

Student Name: Ashish M. Mistari

- Main responsibilities: reasearch, rechargeable model in GaBi
- Specific contributions: rebuild the missing mining processes in GaBi

Student Name: Radwan Kahwaji

- Main responsibilities: reasearch, disposable model in GaBi
- Specific contributions: Retrieving data for inventory

Student Name: Mahmoud Metwaly Aly Elmokadem

- Main responsibilities: reasearch, disposable model in GaBi
- Specific contributions: Scenario Analysis

The four of us worked well together and divided the work fairly. Please note that there was initially a fifth group member. Unfortunately, he had to leave our group because he could not dedicate the necessary time to participate properly in the project. His areas were distributed among the remaining members. Therefore, his contribution to this project amounts to almost nothing. Further information can be found in the email we sent you on December 22nd, and of course also in a personal conversation if desired.

Collaborative Work

The research was conducted individually by each member, but was compiled into a shared file for structuring and further discussion. The two GaBi models were created in a group effort by three members. The report was also a shared document, so that anyone could submit information as needed. The decisions were mainly made during the lectures and group meetings, but some that had to be made quickly were discussed via WhatsApp. Work performance was mainly assessed during the lecture sessions, but could also be accessed from the shared files.

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Appendices

A) GaBi models

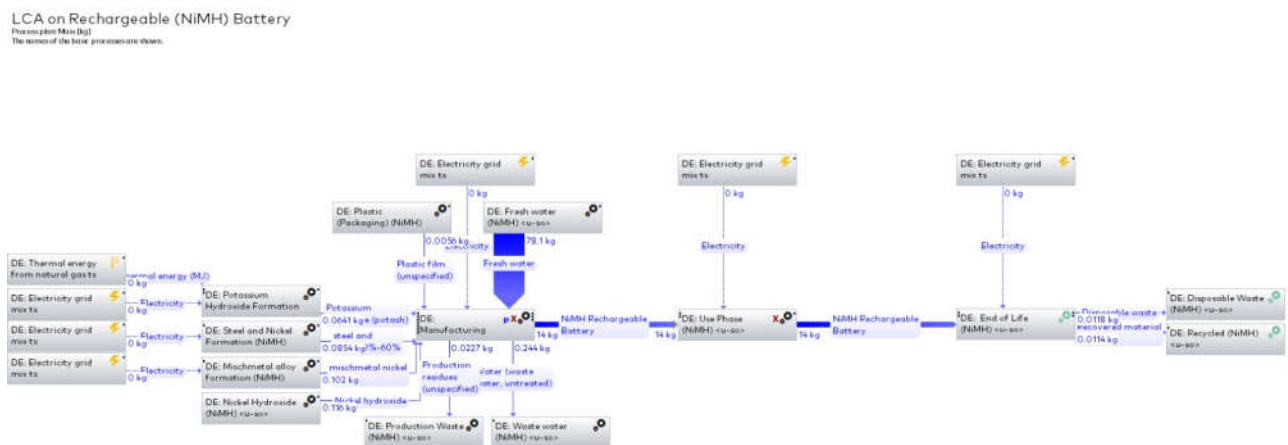


Figure A. 1: GaBi model for the rechargeable batteries

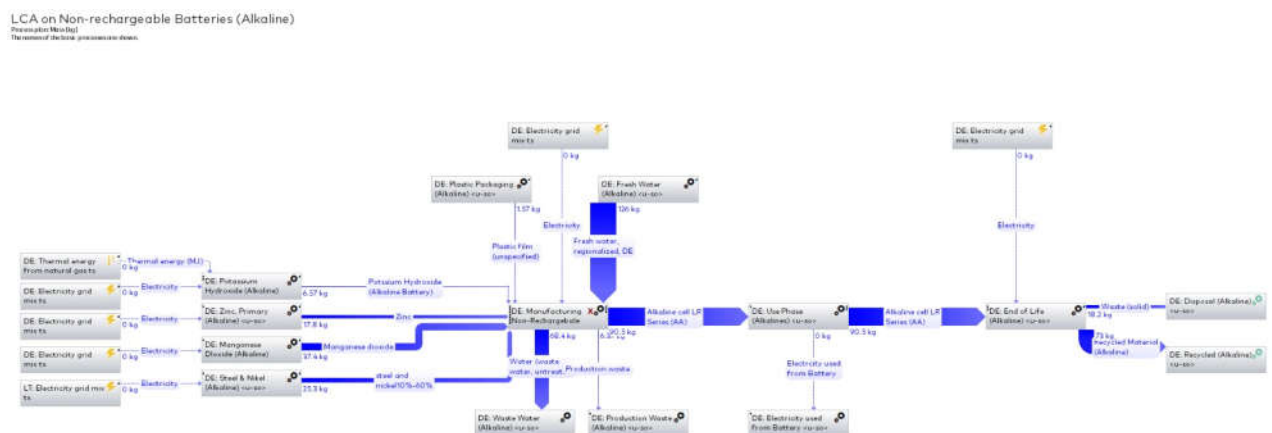


Figure A. 2: GaBi Model for the disposable batteries

B) Scenario Analysis for the Recycling Process

The chosen recycling material should generate high CO₂ emissions during its production from raw materials and be required in large quantities within the battery. Furthermore, recycling should be simple and practical and offer ecological advantages.

Rechargeable Battery Varta AA 2200: Selected Material: Nickel hydroxide Ni(OH)₂

Nickel hydroxide is the main component of rechargeable NiMH batteries. It accounts for about 27% of the total weight of the battery, which is about 8.296 grams per battery. This makes it the largest single component of the battery.

The production of nickel hydroxide from raw materials causes high CO₂ emissions. The process involves the mining, refining, and chemical processing of nickel, all of which are very energy-intensive. The International Energy Agency (IEA) shows that recycling nickel produces much less CO₂ than producing new nickel from ore. Nickel from used NiMH batteries can be recycled and reused to produce new nickel hydroxide for batteries. This means that closed-loop recycling is possible¹⁶¹⁷.

It is assumed that 50% of rechargeable batteries will be collected, which corresponds to current EU collection rates. Of these, 70% of the nickel hydroxide will be recovered during recycling. Some of the material is lost during processing.

The environmental benefits arise from the fact that approximately 2.906 grams of nickel hydroxide can be recovered and reused from a battery. This reduces the demand for new nickel hydroxide by approximately 35% per battery. As a result, mining, energy consumption, and CO₂ emissions are reduced¹⁸¹⁹.

Disposable Battery Varta AA Mignon: Selected material: Steel (battery casing)

Steel is the main metal used in alkaline batteries. It accounts for about 27% of the battery's weight, which is about 6.4 grams per battery.

The production of steel from raw materials causes a lot of CO₂. About 2.2 kg of CO₂ is released per kilogram of new steel. When steel is recycled, CO₂ emissions are significantly lower at around 0.3 kg of CO₂ per kilogram of steel. Steel recycling is

¹⁶ [Flow sheet of the intended investigation on Ni-MH-battery recycling | Download Scientific Diagram](#)

¹⁷ [Recycling of Critical Minerals – Analysis - IEA](#)

¹⁸ [Batteries collection rate | Materials and waste | Circularity Metrics Lab \(EEA\)](#)

¹⁹ [Characterization and leaching of NiCd and NiMH spent batteries for the recovery of metals | Request PDF](#)

simple and well established. It is already carried out worldwide using existing systems, so there is very little technical risk involved²⁰.

It is assumed that 60% of alkaline batteries are collected, with 90% of the containing steel being successfully recycled.

The environmental benefits stem from the fact that approximately 3.5 grams of steel can be recovered from a battery. This reduces the amount of new steel required per battery by around 54%, resulting in significant and direct CO₂ savings²⁰.

Conclusion:

Nickel hydroxide for rechargeable batteries and steel for disposable batteries are extremely suitable materials for recycling. They are used in large quantities, cause high CO₂ emissions during production from raw materials, and can be recycled using proven technologies. Recycling these materials brings real and measurable benefits for the environment.

²⁰ [Life cycle thinking - worldsteel.org](https://worldsteel.org/life-cycle-thinking/)